Using AI Responsibly in Quantitative Social Research

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Research Interest: Using causal and computational methods to evaluate societal and political impacts of digital information technologies

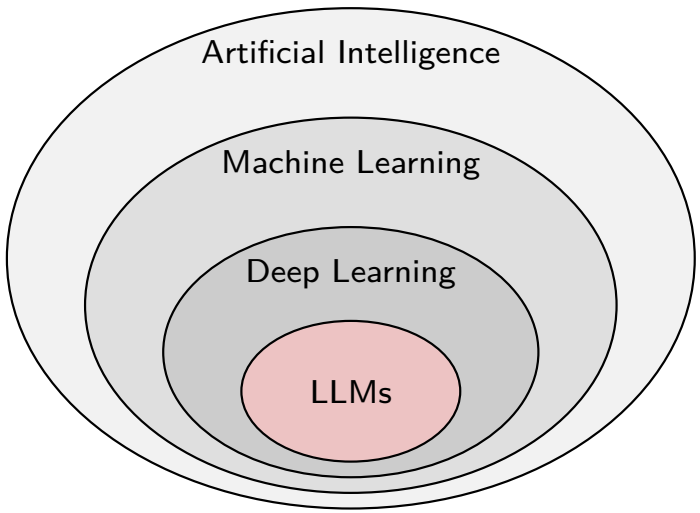
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AI: A Bigger Picture

What Is AI?

- "The science and engineering of making intelligent machines"

  (McCarthy)
- Systems that **perceive** their environment and **act** to achieve goals
- A nested family of techniques, not a single technology



A Brief History of AI

- **1950s** Turing test (1950); "AI" coined at the Dartmouth workshop (1956)
- **1960s–80s** Symbolic AI and expert systems; two "AI winters"
- **1990s–2000s** Statistical machine learning: learning rules from data
- **2012** Deep learning breakthrough in image recognition
- **2017** Transformer architecture
- **2018–2020** Pre-trained language models (BERT, GPT-2, GPT-3)
- **2022–** ChatGPT; **general-purpose LLMs** enter everyday research practice

Pre-LLM AI for Quantitative Research

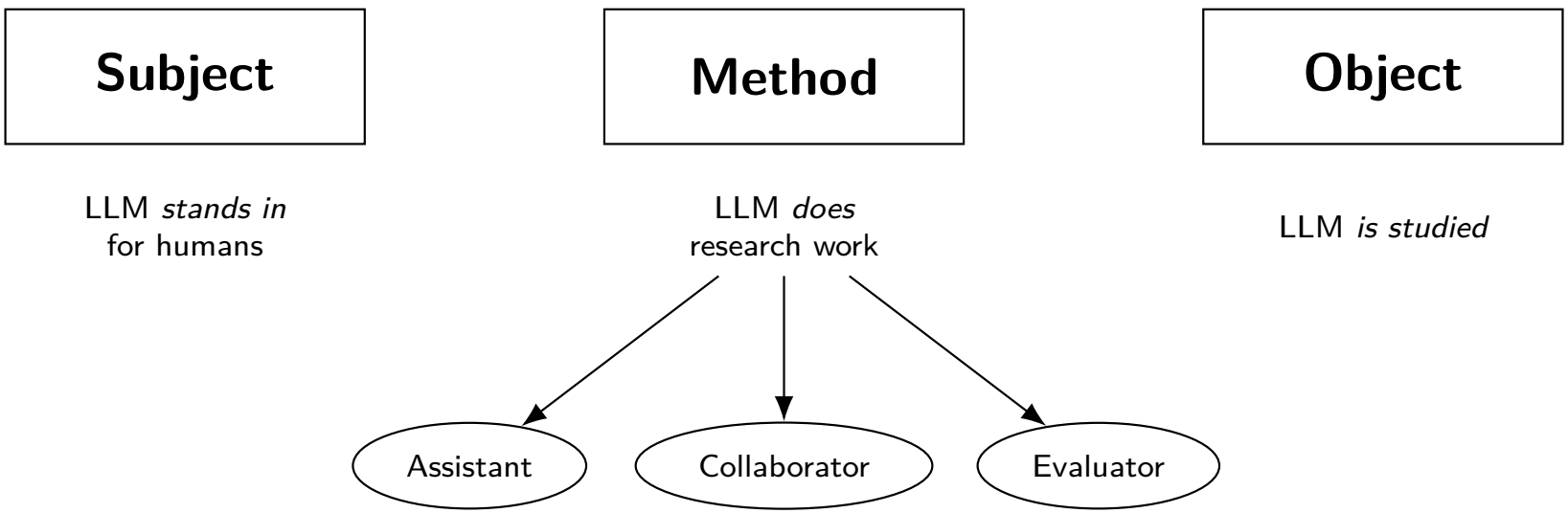
Pre-LLM: Machine Learning as a Measurement Tool

- **Supervised learning**: hand-label a sample, train a classifier, scale up
- **Unsupervised learning**: topic models, clustering, word embeddings
- **Text, images, and networks as data** (King et al., 2013)
- **ML for causal inference**: heterogeneous effects, high-dimensional controls

Common thread: task-specific models, labeled data, and a researcher-built pipeline

LLMs for Quantitative Research

Three Roles of LLMs in Research



LLM as Subject

- **Silicon samples**: simulated survey respondents conditioned on personas
- **Simulated experiments**: pilot designs and power checks before fielding
- **Agent-based simulation**: populations of LLM agents interacting

Key question: when do simulated responses track real human behavior?

LLM as Method: Assistant

- Writing, debugging, and translating analysis code
- Searching and summarizing literature
- Editing and polishing prose
- Data cleaning and wrangling

The researcher specifies the task; the LLM speeds up execution

LLM as Method: Collaborator

- Brainstorming research questions and hypotheses
- Critiquing research designs and identification strategies
- Proposing measures, robustness checks, and alternative explanations
- Agentic workflows: multi-step analysis with researcher oversight

The LLM contributes ideas; the researcher remains accountable

LLM as Method: Evaluator

- **Annotation**: zero- and few-shot text classification at scale
- **LLM-as-judge**: scoring quality, stance, or similarity
- **Review**: feedback on manuscripts, code, and reproducibility

Must be validated against human-coded benchmarks

LLM as Object

- **Auditing models**: political slant, social bias, and value alignment
- **Effects on people**: persuasion, learning, labor, and well-being
- **Effects on information ecosystems**: AI-generated content and search

Limitations

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1. **Validity.** Outputs may not measure what we intend
2. **Reproducibility.** Closed models, silent updates, and stochastic outputs
3. **Bias.** Training data over-represent some groups and languages
4. **Hallucination.** Fabricated facts and citations
5. **Ethics.** Privacy, consent, and data sent to third parties
6. **Over-reliance.** Deskilling and unexamined defaults

Looking Forward

Looking Forward: Responsible Use

- **Disclose**: model, version, prompts, parameters, and date
- **Validate**: compare against human benchmarks before scaling
- **Reproduce**: pin versions; prefer open-weight models when possible
- **Stay accountable**: humans own the design, inference, and claims

From AI as a tool to AI as a research partner—with guardrails

Acknowledgments

We thank .... All errors are our own.

References

References I

King, G., Pan, J., & Roberts, M. E. (2013). How censorship in China allows government criticism but silences collective expression. *American Political Science Review*, *107*(2), 326–343. https://doi.org/10.1017/S0003055413000014